Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	23.27 final answer	**2**	**M1** for 9 × 2.97 soi
1(a)(ii)	2.75 final answer	**3**	**M2** for $2.97 \div \frac{108}{100}$ oe or **M1** for 108[%] associated with 2.97 oe
1(b)	12.4[0] or 12.41 to 12.42	**2**	**M1** for 35 ÷ 0.0153 oe If 0 scored, **SC1** for answer 0.19
1(c)	70 nfww	**3**	**M2** for (600 + 2.5) ÷ (9 – 0.5) or **B1** for one of 600 + 2.5 or 9 – 0.5 seen
2(a)	128	**2**	**M1** for $4 \times \frac{1}{2} \times 8 \times 8$ oe
2(b)(i)	18.3 or 18.26 to 18.29…	**3**	**M2** for $\frac{1}{4}(\pi \times 8^2 - \textit{their}\,128)$ oe or **M1** for $\pi \times 8^2 - \textit{their}\,128$ oe or for $\frac{1}{4} \times \pi \times 8^2$ oe OR **SC2dep** for answer 4.56 to 4.57...
2(b)(ii)	23.9 or 23.87 to 23.882	**4**	**M3** for $\frac{90}{360} \times 2 \times \pi \times 8 + \sqrt{8^2 + 8^2}$ oe OR **M1** for $\frac{90}{360} \times 2 \times \pi \times 8$ oe **M1** for $\sqrt{128}$ oe OR **SC3dep** for answer 11.9 or 11.93 to 11.94…
3(a)	0 –0.17 2.4	**3**	**B1** for each
3(b)	Fully correct smooth curve	**4**	**B3FT** for 9 or 10 correct points or **B2FT** for 7 or 8 correct points or **B1FT** for 5 or 6 correct points
3(c)	$x \leqslant 0.17$ to 0.25 and $x \geqslant 2.25$ to 2.3	**3**	**B2** for strict inequalities or one correct or **B1** for 0.17 to 0.25 and 2.25 to 2.3 seen

Question	Answer	Marks	Partial Marks
3(d)(i)	$y = 4 - x$ oe final answer	**2**	**B1** for $4 - x$ or $y = k - x$ or $y = 4 + kx$ oe
3(d)(ii)	correct ruled line	**1**	**FT** if in form $y = mx + c$ oe ($m, c \neq 0$)
	0.125 to 0.2 and 2.15 to 2.2	**2**	**B1** for each
4(a)	$[\pm]\sqrt{k - s}$ final answer	**2**	**M1** for $t^2 = k - s$
4(b)(i)	$(x - 5)(x + 5)$ final answer	**1**	
4(b)(ii)	$\frac{x-5}{x-7}$ nfww final answer	**3**	**M2** for $(x - 7)(x + 5)$ or **M1** for $x(x + 5) - 7(x + 5)$ or $x(x - 7) + 5(x - 7)$ or $(x + a)(x + b)$ where $a + b = -2$ or $ab = -35$
4(c)	$\frac{4x^2 - 7x - 8}{x(x+1)}$ or $\frac{4x^2 - 7x - 8}{x^2 + x}$ final answer	**3**	**M1** for $(x - 8)(x + 1) + 3x \times x$ oe isw **B1** for common denominator $x(x + 1)$ oe isw
4(d)	3, 4, 5, 6 nfww	**3**	**B2** for 3 correct or 4 correct and 1 extra or **M2** for $n > \frac{18}{8}$ oe and $n \leqslant 6$ or **M1** for $18 < 8n$ [$\leqslant 30 + 3n$] or [$18 - 3n <$] $5n \leqslant 30$ seen
5(a)(i)	1930 or 1940 or 1933.4 to 1935.3	**5**	**B1** for interior angle 120 soi or angle at centre 60 soi or for correct use of Pythagoras' with 7 and 3.5 or with 14 and 7 **M3** for $6 \times \frac{1}{2} \times 7^2 \times \sin 60 \times 15.2$ oe or complete other methods or **M2** for $6 \times \frac{1}{2} \times 7^2 \times \sin 60$ oe OR **M1** for $\frac{1}{2} \times 7^2 \times \sin 60$ oe or other partial area of hexagon **M1dep** for *their* area × 15.2 evaluated